AGV I-PRIME driverless towing vehicle

Circuit diagram

Document number IGP-IPR-CD-001 · Revision A · Language en

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Documents that belong with this circuit diagram

Document	Number
Operation manual	IGP-IPR-OM-001
Component list	IGP-IPR-CL-001

Revision history

Rev	Change	Date	Name
A	First issue, draft	2026-11-01	JOY M.

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A	First issue, draft	2026-11-01	JOY M.

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Title: AGV I-PRIME driverless towing vehicle
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Page	Description	Location	Status
1	Cover		Rev A
2	Contents		Rev A
4	Symbol legend		Rev A
10	Overview: power distribution (single line)		Rev A
11	Overview: communication (CAN, Ethernet)		Rev A
12	Overview: safety chain		Rev A
20	Battery, main breaker, 48 V distribution	+BT +EB	Rev A
21	Motor drivers: power and motors	+EB +VH	Rev A
30	24 V control supply, part 1	+EB	Rev A
31	24 V control supply, part 2	+EB	Rev A
32	12 V supply	+EB	Rev A
40	Safety: emergency stop, mode selector, reset	+EB +FR	Rev A
41	Safety: safety laser scanner, front	+EB +VH	Rev A
42	Safety: safe torque off and feedback (EDM)	+EB	Rev A
43	Safety controller: supply and modules	+EB	Rev A
44	Safety controller I/O table	+EB	Rev A
45	Safety: wheel encoders and drive monitor	+EB +VH	Rev A
50	Motor drivers: control connector	+EB	Rev A
51	CAN bus	+EB +VH	Rev A
60	I/O module: inputs	+EB +FR +PD	Rev A
61	I/O module: outputs	+EB +VH	Rev A
62	I/O module I/O table	+EB	Rev A
75	Cable overview		Rev A
80	Parts list		Rev A

Page numbers are grouped by subject, so the gaps are intentional: 1–9 front matter, 10–19 overview, 20–29 48 V power, 30–39 control supplies, 40–49 safety, 50–59 drives, 60–69 control I/O, 70–79 cables, 80–89 lists.

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0 1 2 3 4 5 6 7 8 9

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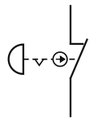
Location:
Language: en

Document number: IGP-IPR-CD-001
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Make contact (NO)

Terminals 13-14, 23-24. Drawn in rest position, supplies off



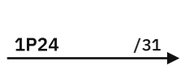
Emergency stop

Red mushroom head, latching, turn to release



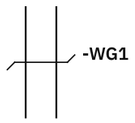
Wire marker

Potential or signal name printed on the wire marker, written along the wire



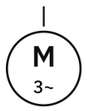
Potential rail

Name and source left, continuation arrow and target right



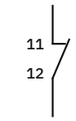
Cable

Line across the conductors of one cable, with its tag and type



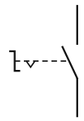
Motor

Three-phase motor (brushless DC drive)



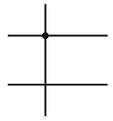
Break contact (NC)

Terminals 11-12, 21-22



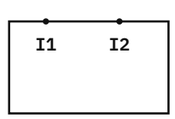
Selector switch, maintained

Turn operator, latching. Position table beside the switch: X = contact closed



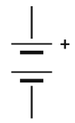
Junction / crossing

Dot = connected. Crossing without a dot = not connected



Module or device

Box with the terminals in use on its edge; tag and type inside



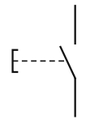
Battery

\+ at the long plate



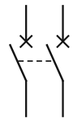
Protective earth

Chassis bonding point



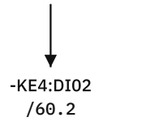
Push-button, momentary

Push operator, returns by spring



Circuit breaker

Poles of one breaker linked by a dashed line



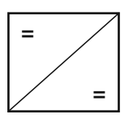
Interruption point

Continues at the target shown. /41.3 = page 41, column 3



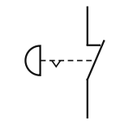
Location box (dash-dot)

Devices inside are in the location written at its top-left corner



DC/DC converter, motor driver

Box with diagonal: input left, output right (= direct current, ~ alternating current)



Push-button, latching

Mushroom head, turn to release. Not an emergency stop (black)

Locations (+)

Code	Location
+EB	Electrical box (harness board): breakers, converters, drivers, controllers
+FR	Front control panel: emergency stop, push-buttons, selector
+VH	Vehicle body and chassis: motors, scanner, line sensor, horn
+BT	Battery compartment
+PD	Pendant (hand-held, wired)

Potentials

Name	Meaning
1P48 / 1N48	48 V bus after -QA1, + / return. Floating: not connected to the chassis
2P48, 3P48	48 V feeders after -FC1 (left driver), -FC2 (right driver)
1P24 / 1N24	24 V control bus after -FC3
1P12 / 1N12	12 V bus after -FC4. 1N12 is joined to 1N24
PE	Chassis / protective bonding of metal parts

Product classes (-), IEC 81346-2

Class	Meaning	Tags in this set
BG	Sensing position or presence (scanner, line sensor, RFID reader, encoders)	-BG1, -BG3 … -BG6
FC	Protecting against overcurrent (circuit breakers)	-FC1 … -FC4
GB	Storing electrical energy (battery)	-GB1
GQ	Moving gases (fans)	-GQ1 … -GQ3
KE	Processing signals (safety controller, I/O modules)	-KE1 … -KE5
MA	Driving by electric motor	-MA1, -MA2
PH	Presenting information visually (display)	-PH1
PJ	Presenting information acoustically (horn)	-PJ1
QA	Switching the main power supply (supply disconnecting device)	-QA1
SJ	Manual control (emergency stop, push-buttons, selector)	-SJ1 … -SJ9
TA	Converting electrical energy (DC/DC converters, motor drivers)	-TA1 … -TA4
TF	Converting signals (USB-to-CAN interface)	-TF1
XE	Connecting to earth (protective earth point)	-XE1
WD / WG	Cables: power / control and data	-WD1 … -WD12; -WG1 … -WG13

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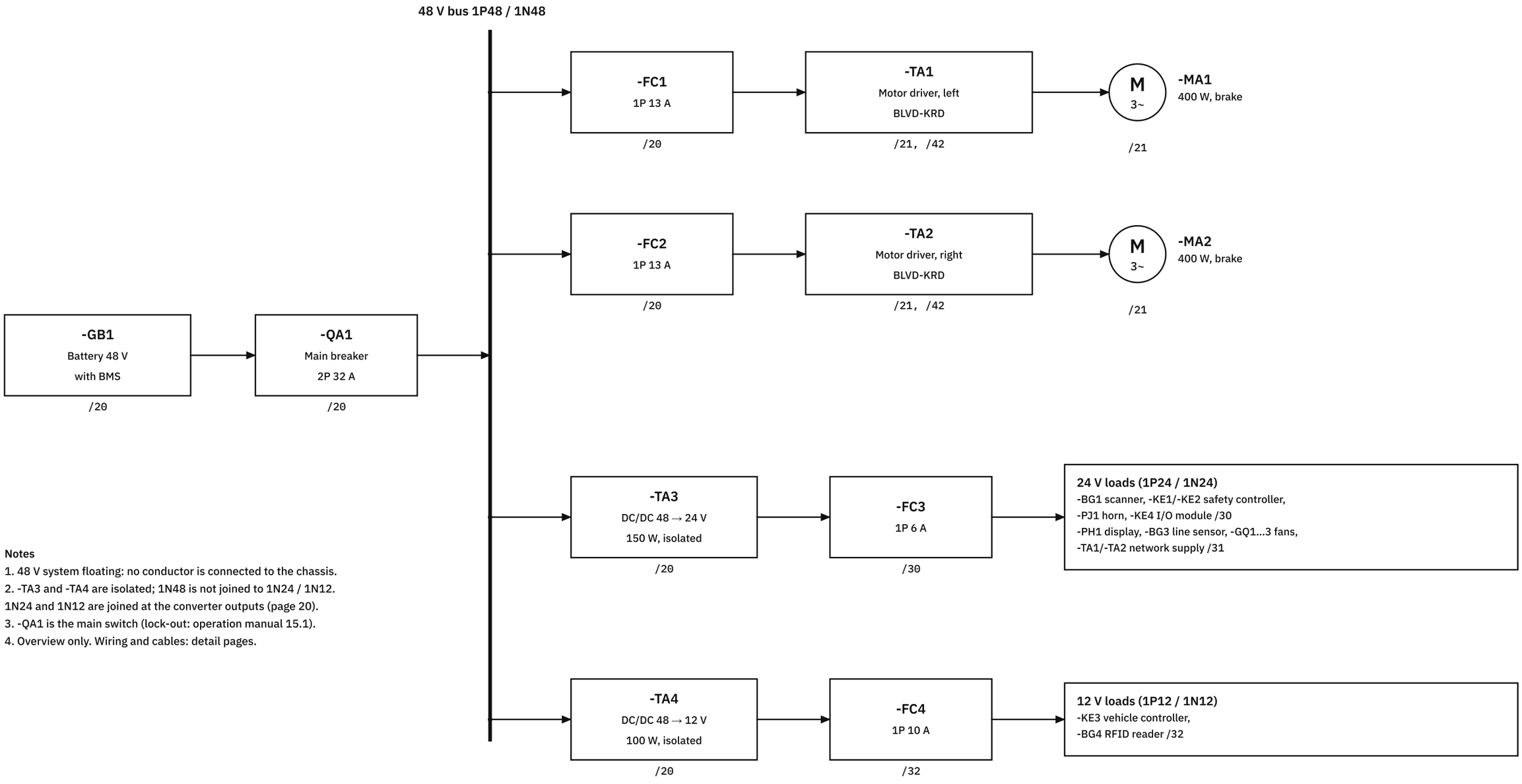
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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **Symbol legend**

Location:
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 4
Number of pages: 24

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Notes

1. 48 V system floating: no conductor is connected to the chassis.
2. -TA3 and -TA4 are isolated; 1N48 is not joined to 1N24 / 1N12. 1N24 and 1N12 are joined at the converter outputs (page 20).
3. -QA1 is the main switch (lock-out: operation manual 15.1).
4. Overview only. Wiring and cables: detail pages.

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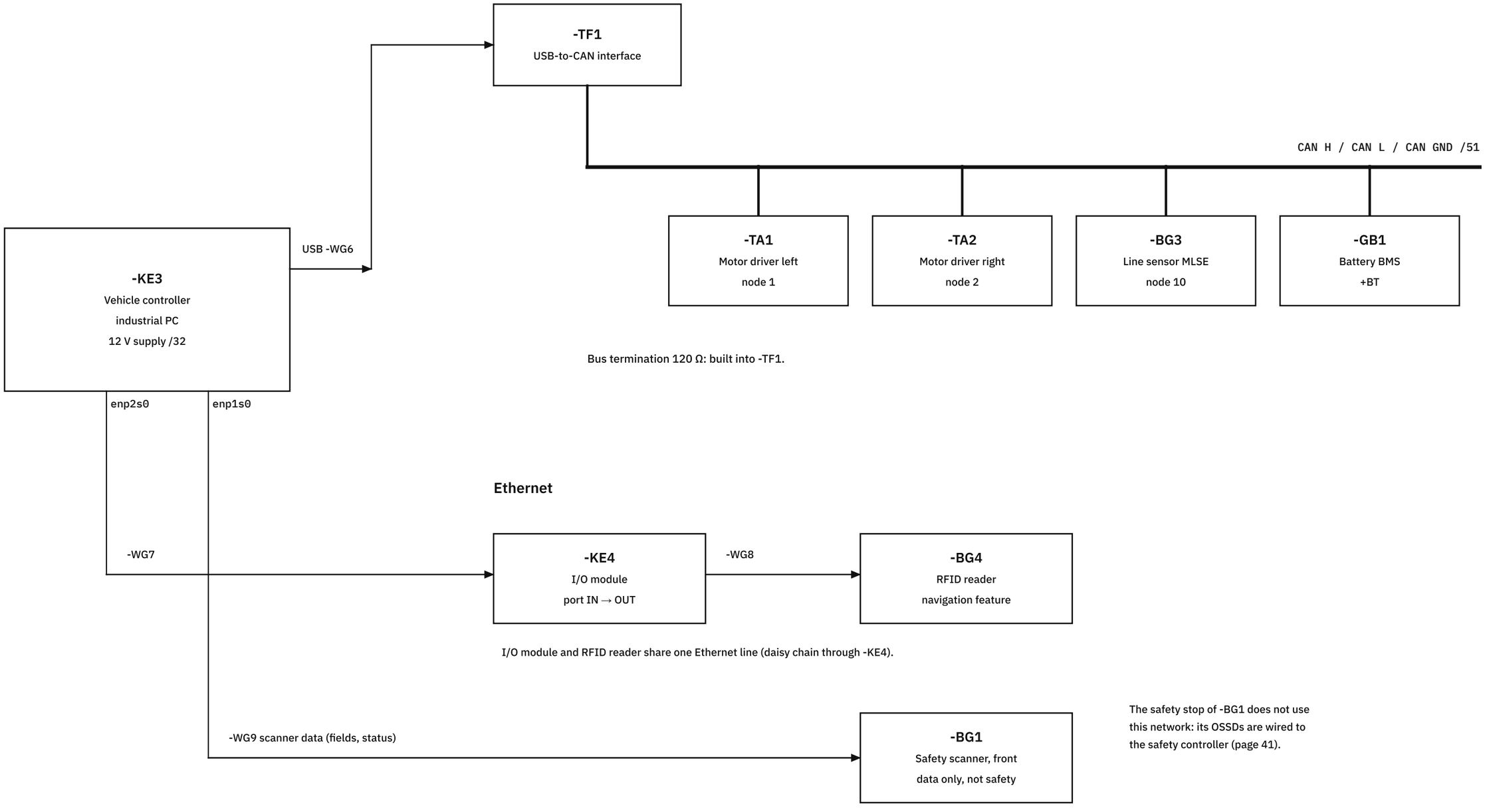
Page description: **Overview: power distribution (single line)**

Location:

Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: **10**
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CAN bus, 125 kbit/s, CANopen



Bus termination 120 Ω: built into -TF1.

I/O module and RFID reader share one Ethernet line (daisy chain through -KE4).

The safety stop of -BG1 does not use this network: its OSSDs are wired to the safety controller (page 41).

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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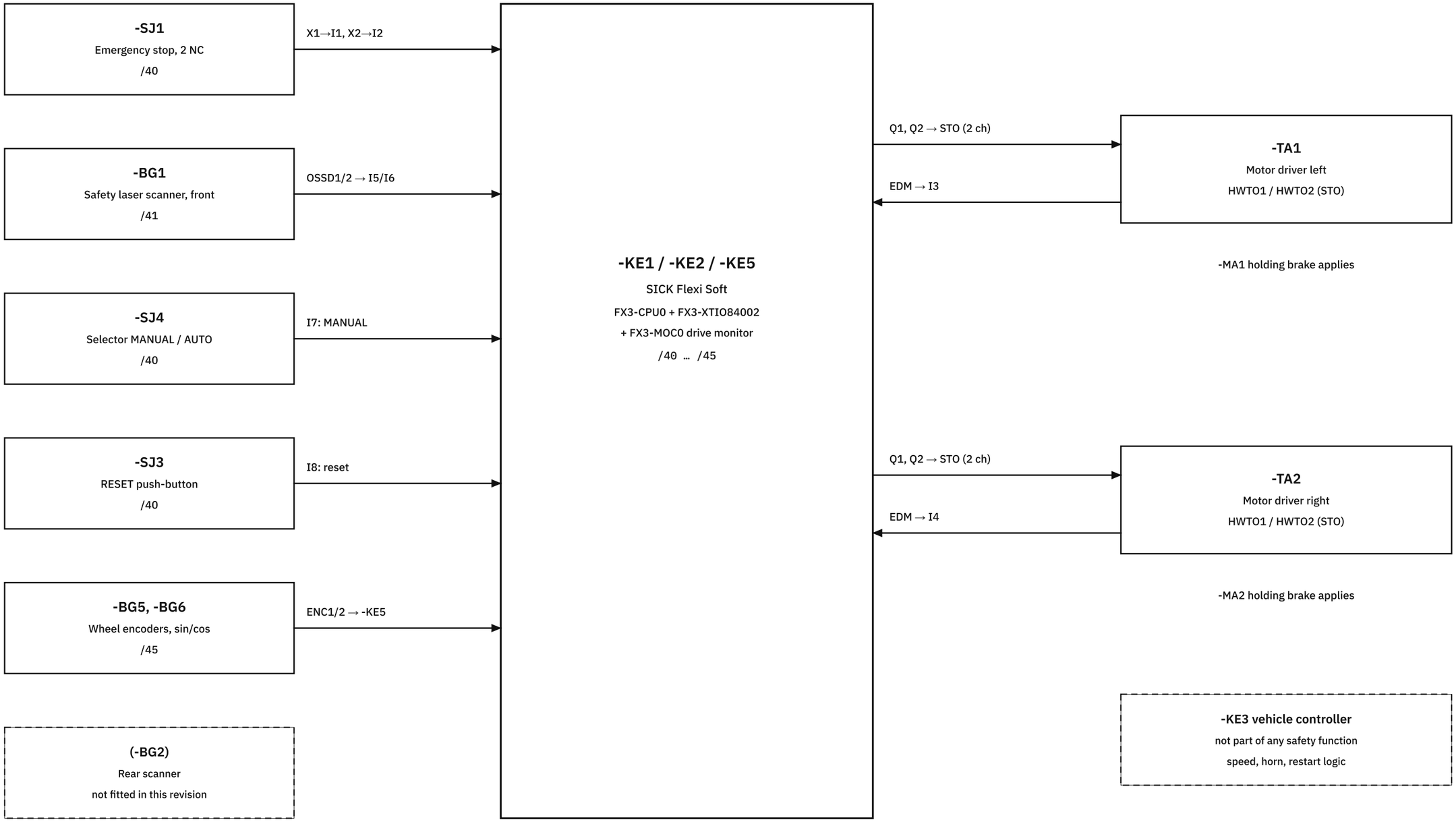
Date of issue: 2026-11-01
Created by: JOY M.
Checked by: JOKI I.
Approved by: ROBERTA A.

Title: AGV I-PRIME driverless towing vehicle
Document type: Circuit diagram
Page description: Overview: communication (CAN, Ethernet)

Location:
Language: en

Document number: IGP-IPR-CD-001
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Stop: torque removed on both drivers (STO), holding brakes apply.

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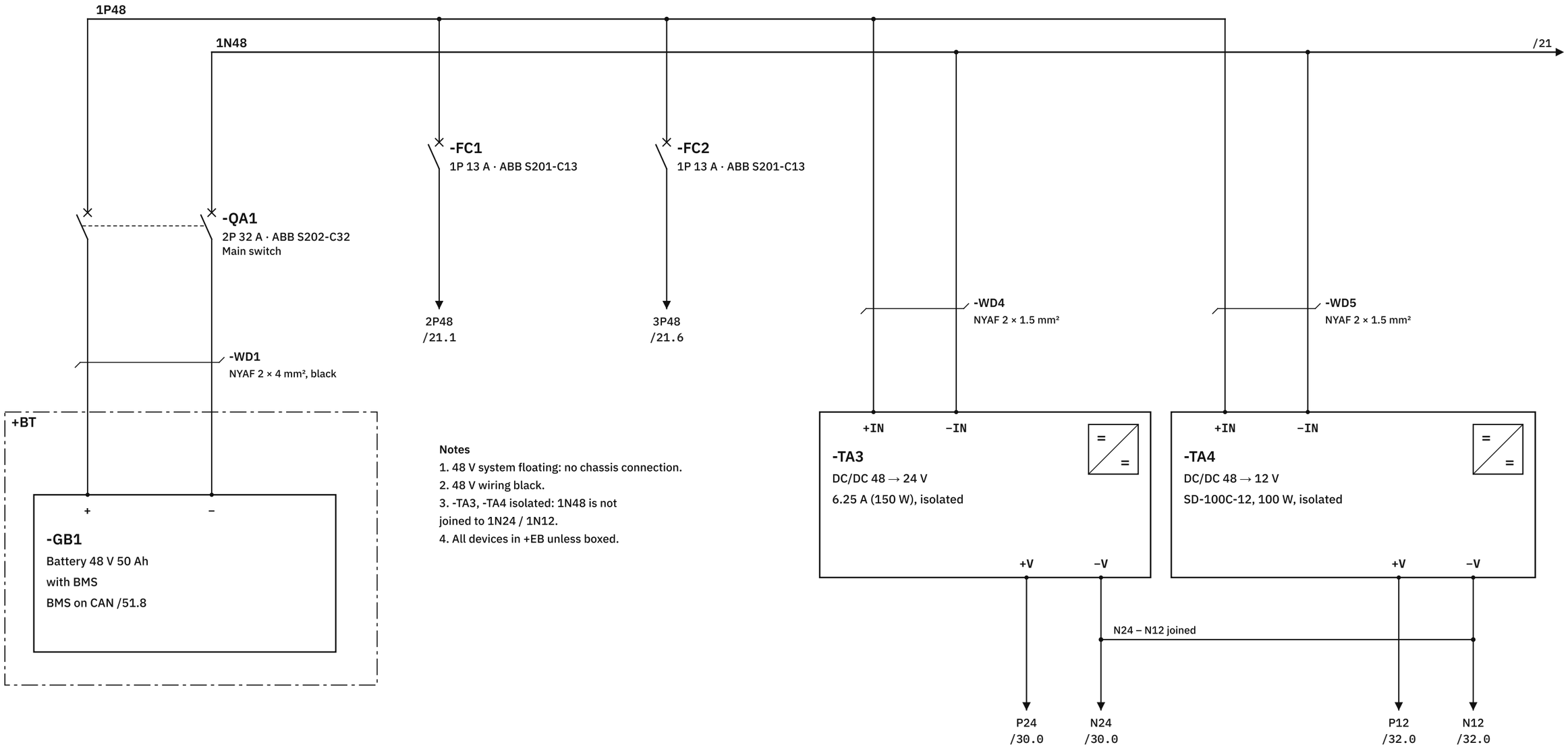
Document type: Circuit diagram

Page description: **Overview: safety chain**

Location:

Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: **12**
Number of pages: 24



Notes

1. 48 V system floating: no chassis connection.
2. 48 V wiring black.
3. -TA3, -TA4 isolated: 1N48 is not joined to 1N24 / 1N12.
4. All devices in +EB unless boxed.

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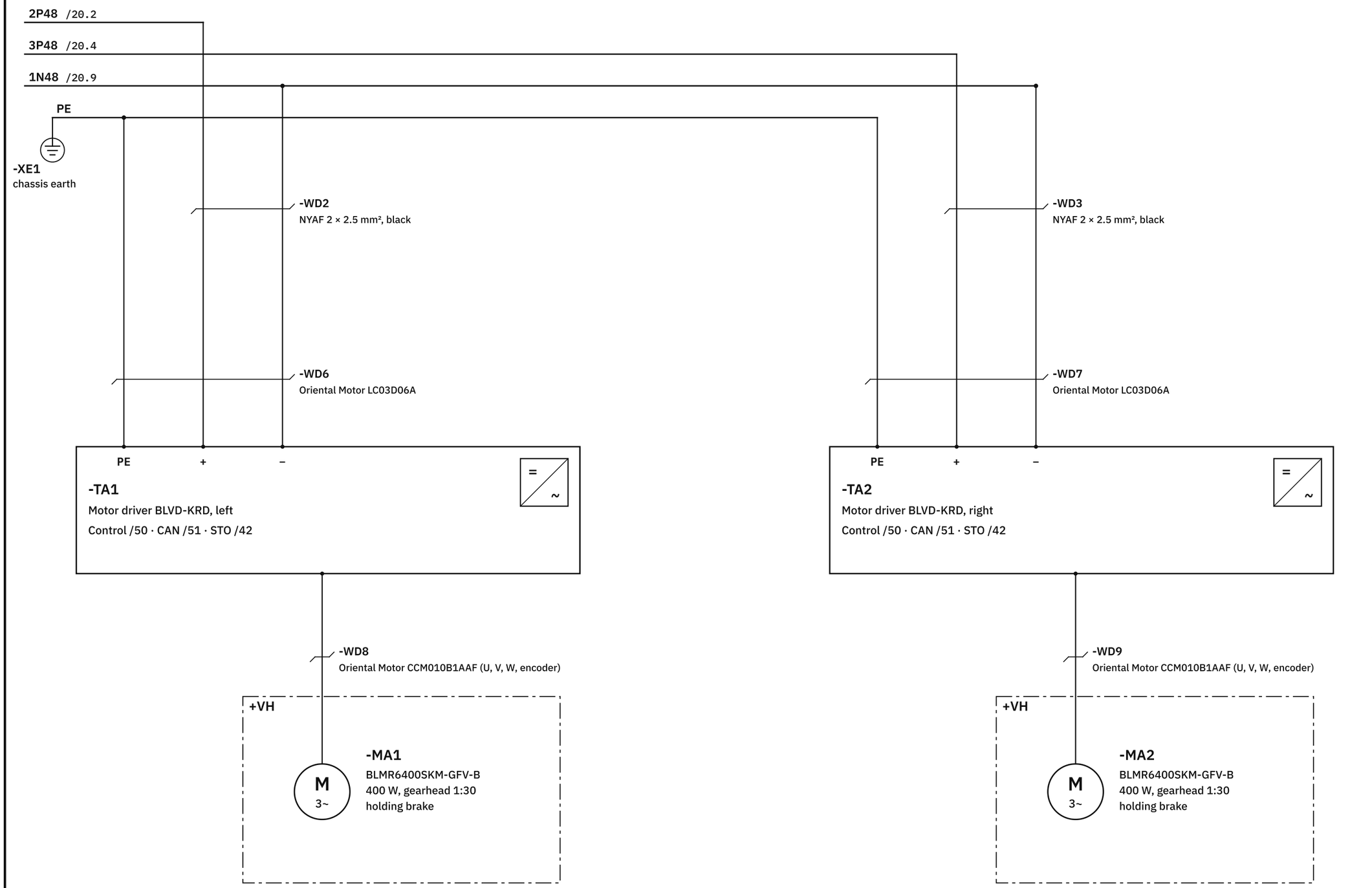
Page description: **Battery, main breaker, 48 V distribution**

Location: +BT +EB

Language: en

Document number: IGP-IPR-CD-001
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Page: **20**
Number of pages: 24

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Notes

1. 48 V wiring black.
2. PE: bonding of the driver housings to the chassis.
3. Motor and brake are fed by the driver. All their power comes in on -WD6 / -WD7 (LC03, 48 V).

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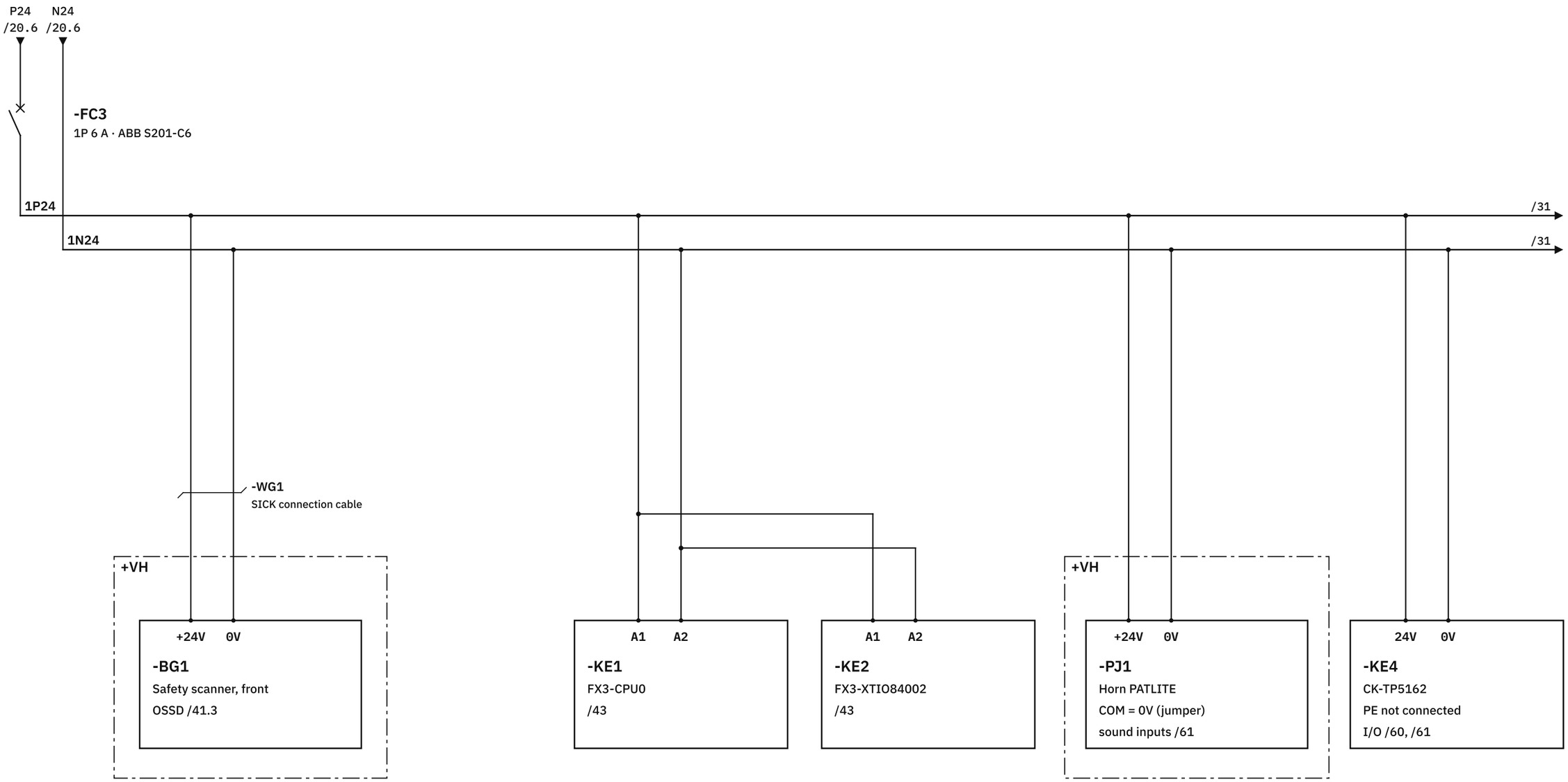
Page description: **Motor drivers: power and motors**

Location: +EB +VH

Language: en

Document number: IGP-IPR-CD-001
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Notes

1. Control wiring AWG22 (0.34 mm²) blue unless marked.
2. Rear scanner not fitted in this revision.
3. All devices in +EB unless boxed.

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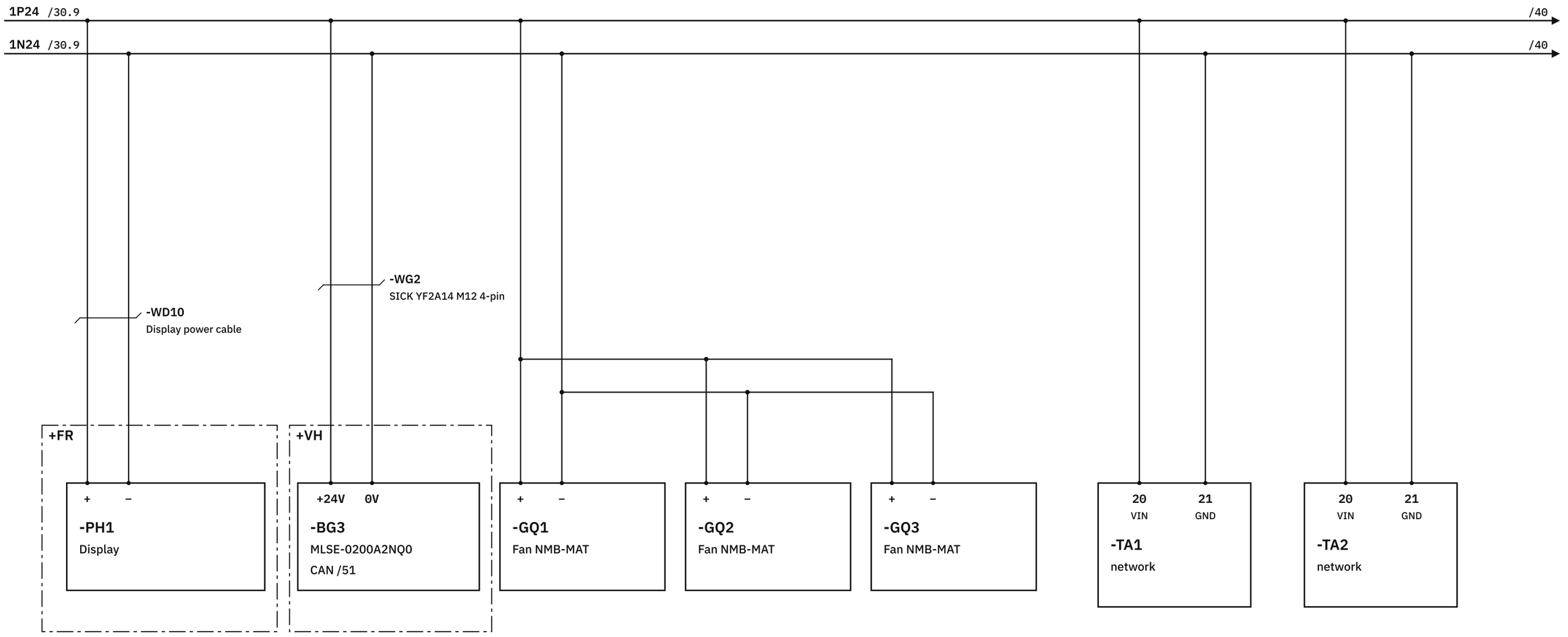
Date of issue: 2026-11-01
Created by: JOY M.
Checked by: JOKI I.
Approved by: ROBERTA A.

Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **24 V control supply, part 1**

Location: +EB
Language: en

Document number: IGP-IPR-CD-001
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Notes

1. Control wiring AWG22 (0.34 mm²) blue unless marked.
2. -BG3 line sensor: the same cable carries CAN (page 51).
3. -TA1, -TA2: network supply only. Driver power: page 21.
4. All devices in +EB unless boxed.

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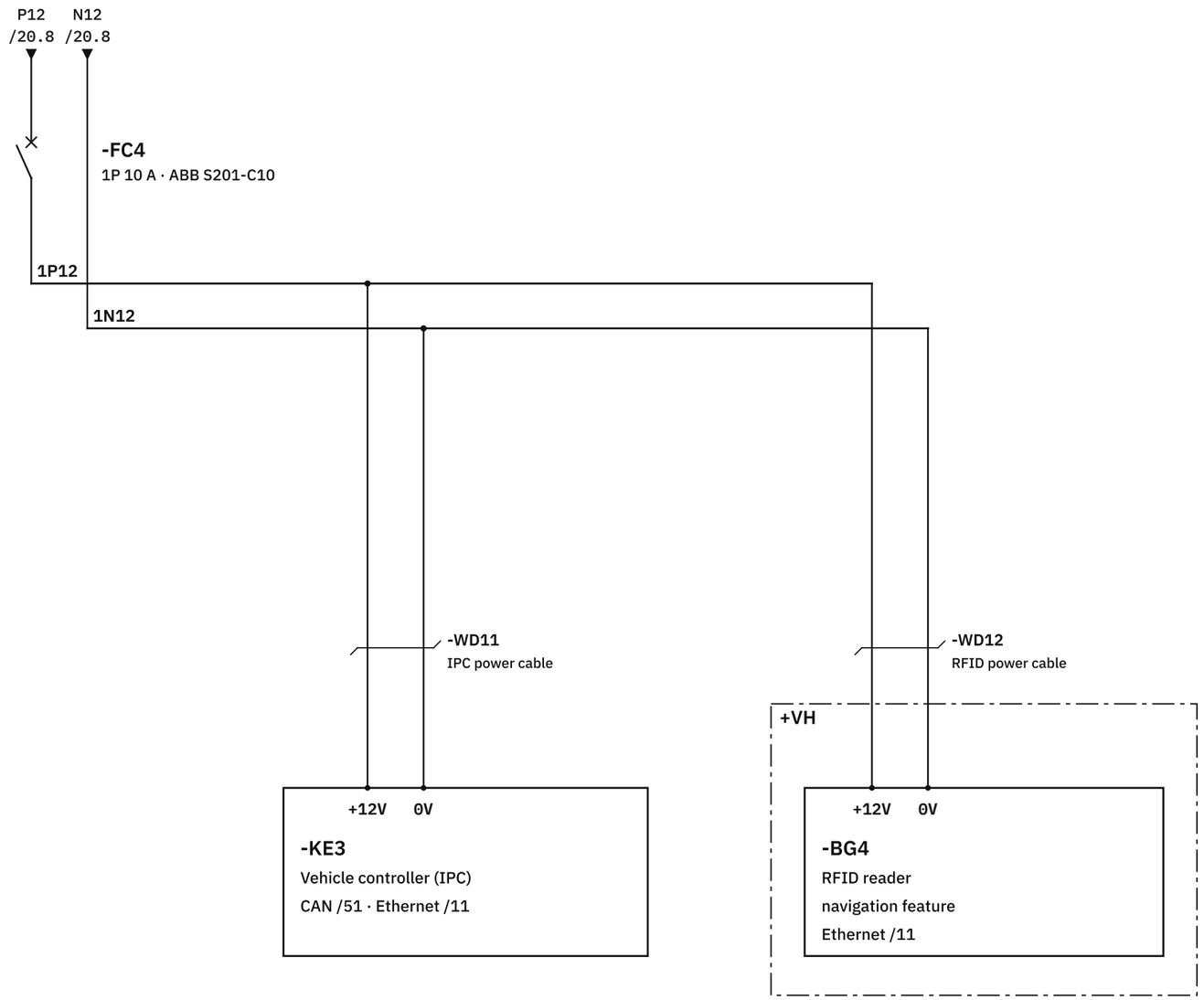
Document type: Circuit diagram

Page description: **24 V control supply, part 2**

Location: +EB

Language: en

Document number: IGP-IPR-CD-001
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Page: **31**
Number of pages: 24



Notes

1. 1N12 is joined to 1N24 at the converter outputs (page 20).
2. Control wiring AWG22 (0.34 mm²) blue.
3. All devices in +EB unless boxed.

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram

Page description: **12 V supply**

Location: +EB

Language: en

Document number: IGP-IPR-CD-001
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1P24 /31.9 → /42

1N24 /31.9 → /42

Notes

1. Contacts drawn in rest position, supplies off. -SJ4 drawn in AUTO (contacts open).
2. X1, X2 are pulsed test outputs of -KE2. X1 → I1 and X2 → I2.
3. Wiring AWG22 (0.34 mm²) blue unless stated.

+FR front panel

-SJ1

Emergency stop
2 NC
turn to release
ARE-3-R-22-R

11 12 — 21 22

+FR front panel

-SJ4

Selector MANUAL / AUTO
2 NO, maintained
ARS-S-2-12

Pos.	13-14	23-24
MANUAL	X	X
AUTO		

13 14 — 23 24

+FR front panel

-SJ3

RESET push-button, yellow
2 NO, momentary
ARF-F-1-20-Y

13 14 — 23 24

-KE4:DI02 /60.2

-KE4:DI01 /60.1

+EB electrical box

X1 I1 X2 I2 I7 I8

-KE2

SICK Flexi Soft FX3-XTIO84002 · safety controller I/O module (main module -KE1)
Other terminals: A1, A2 /43 · I3, I4 /42 · I5, I6 /41 · Q1, Q2 /42 · Q3, Q4 spare

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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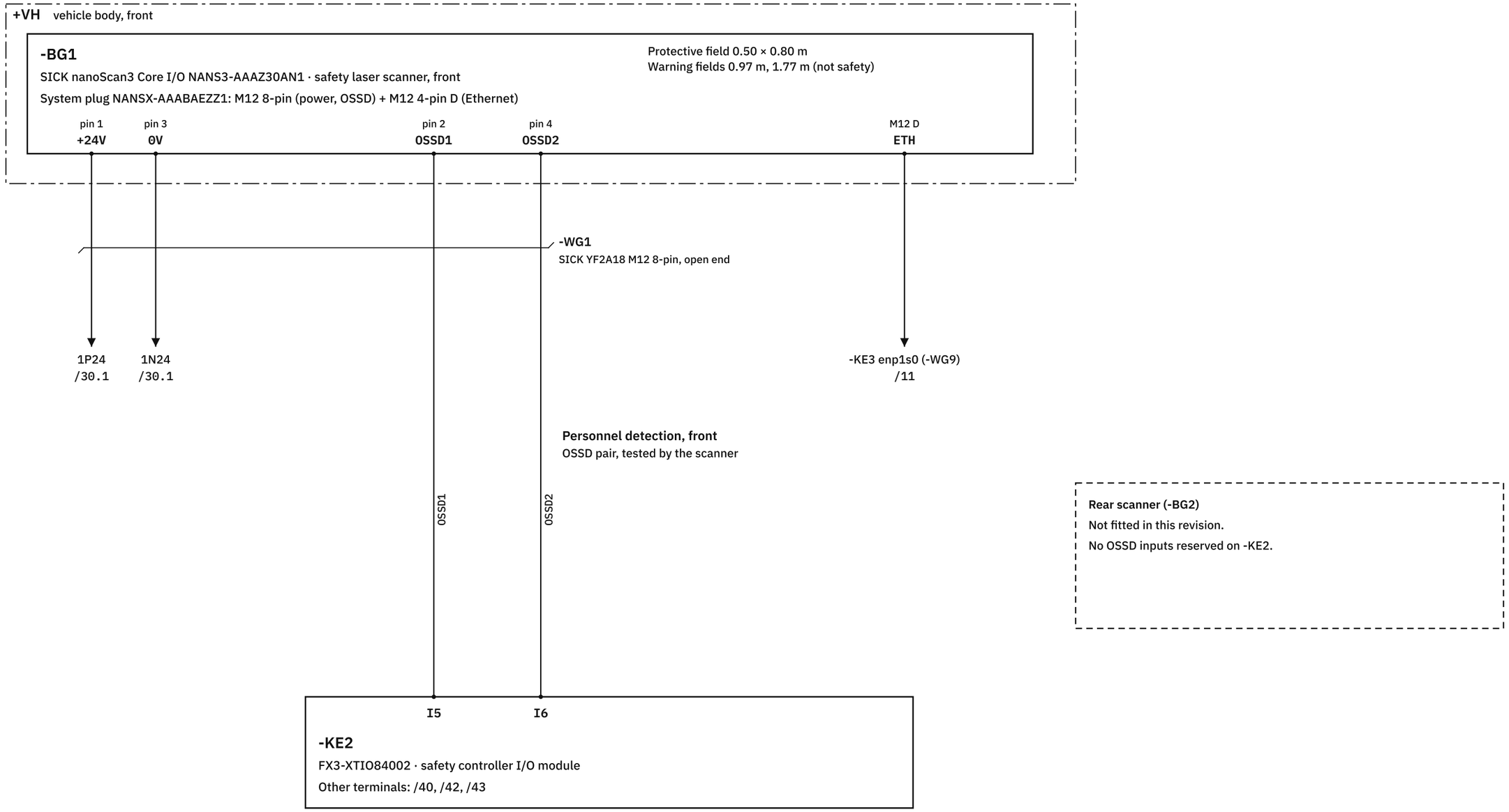
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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **Safety: emergency stop, mode selector, reset**

Location: +EB +FR
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 40
Number of pages: 24

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Notes

1. -KE2 inputs I5 / I6: one OSSD input pair.
2. The Ethernet link carries field and status data for the vehicle controller only (page 11).

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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **Safety: safety laser scanner, front**

Location: +EB +VH
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: **41**
Number of pages: 24

1P24 /40 → /60

1N24 /40 → /60

-KE2
FX3-XTIO84002

Q1 · Q2 · I3 · I4

HWTO1 · HWTO2 · EDM-L · EDM-R

Notes

1. Q1 and Q2 switch off → torque removed on both drivers (STO).
2. EDM: -KE2 checks that torque is really off (I3 left, I4 right).
3. HWTO1 and HWTO2 are looped from -TA1 to -TA2 (junctions).
4. Q3, Q4 spare.
5. Wiring AWG22 (0.34 mm²) blue. All devices in +EB.

-TA1
Motor driver BLVD-KRD, left · control connector (page 50)
Factory jumpers 11–25, 12–26, 13–27 removed

11	26	14	12	27	28
HWTO1+	HWTO2+	EDM–	HWTO1–	HWTO2–	EDM+

-TA2
Motor driver BLVD-KRD, right · control connector (page 50)
Factory jumpers 11–25, 12–26, 13–27 removed

11	26	14	12	27	28
HWTO1+	HWTO2+	EDM–	HWTO1–	HWTO2–	EDM+

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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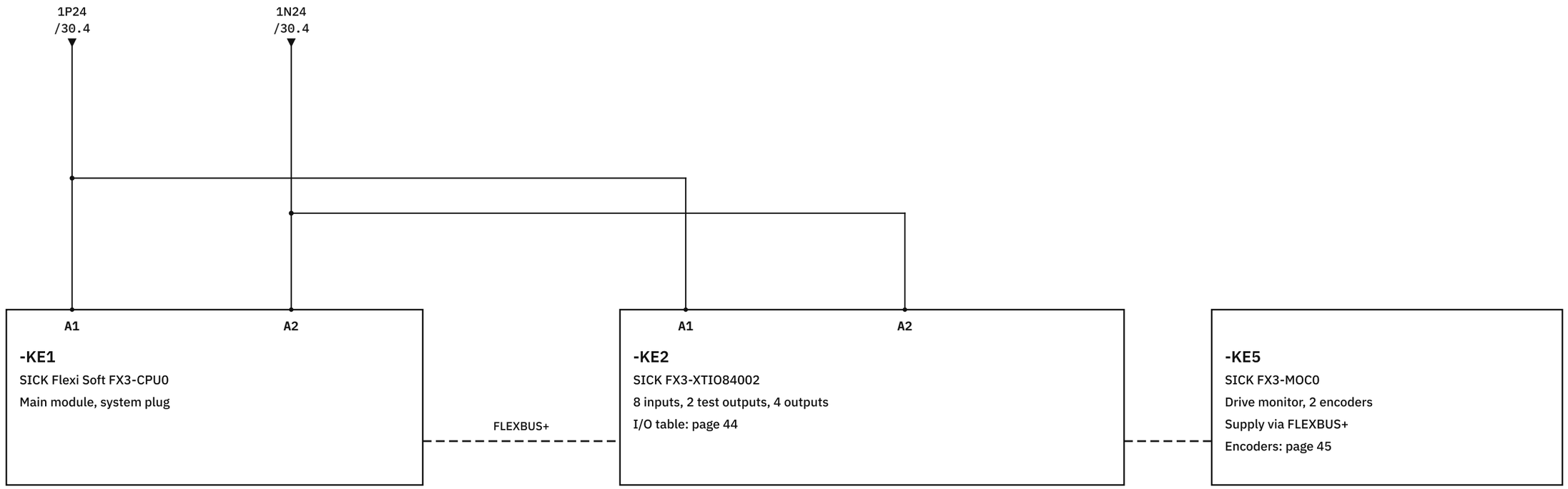
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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **Safety: safe torque off and feedback (EDM)**

Location: +EB
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 42
Number of pages: 24

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-KE2 terminals: inputs

Terminal	Connected to	Page
X1	-SJ1 (11-12) → I1	
X2	-SJ1 (21-22) → I2	
I1	-SJ1 (11-12)	
I2	-SJ1 (21-22)	
I3	-TA1:14 EDM–	
I4	-TA2:14 EDM–	
I5	-BG1 front scanner	
I6	-BG1 front scanner	
I7	-SJ4 (13-14), closed in MANUAL	
I8	-SJ3 (13-14)	

-KE2 terminals: supply and outputs

Terminal	Connected to	Page
A1	supply	
A2	supply	
Q1	-TA1:11, -TA2:11 HWTO1+	
Q2	-TA1:26, -TA2:26 HWTO2+	
Q3	spare	
Q4	spare	

Notes

1. After any change of configuration or wiring: verify all safety functions (operation manual 8.5).
2. All devices in +EB.

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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **Safety controller: supply and modules**

Location: +EB
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 43
Number of pages: 24

-KE2 SICK FX3-XTIO84002 — safety controller I/O module (main module -KE1 FX3-CPU0)

Terminal	Signal	Device / connection	Test pulse	Page
A1	1P24	supply	—	43
A2	1N24	supply	—	43
X1	test output 1	-SJ1 (11-12) → I1	source	40
X2	test output 2	-SJ1 (21-22) → I2	source	40
I1	E-stop channel 1	-SJ1 (11-12)	X1	40
I2	E-stop channel 2	-SJ1 (21-22)	X2	40
I3	EDM left	-TA1:14 EDM–		42
I4	EDM right	-TA2:14 EDM–		42
I5	OSSD 1	-BG1 front scanner	scanner-internal	41
I6	OSSD 2	-BG1 front scanner	scanner-internal	41
I7	MANUAL	-SJ4 (13-14), closed in MANUAL		40
I8	RESET	-SJ3 (13-14)		40
Q1	STO channel 1	-TA1:11, -TA2:11 HWTO1+		42
Q2	STO channel 2	-TA1:26, -TA2:26 HWTO2+		42
Q3		spare		
Q4		spare		

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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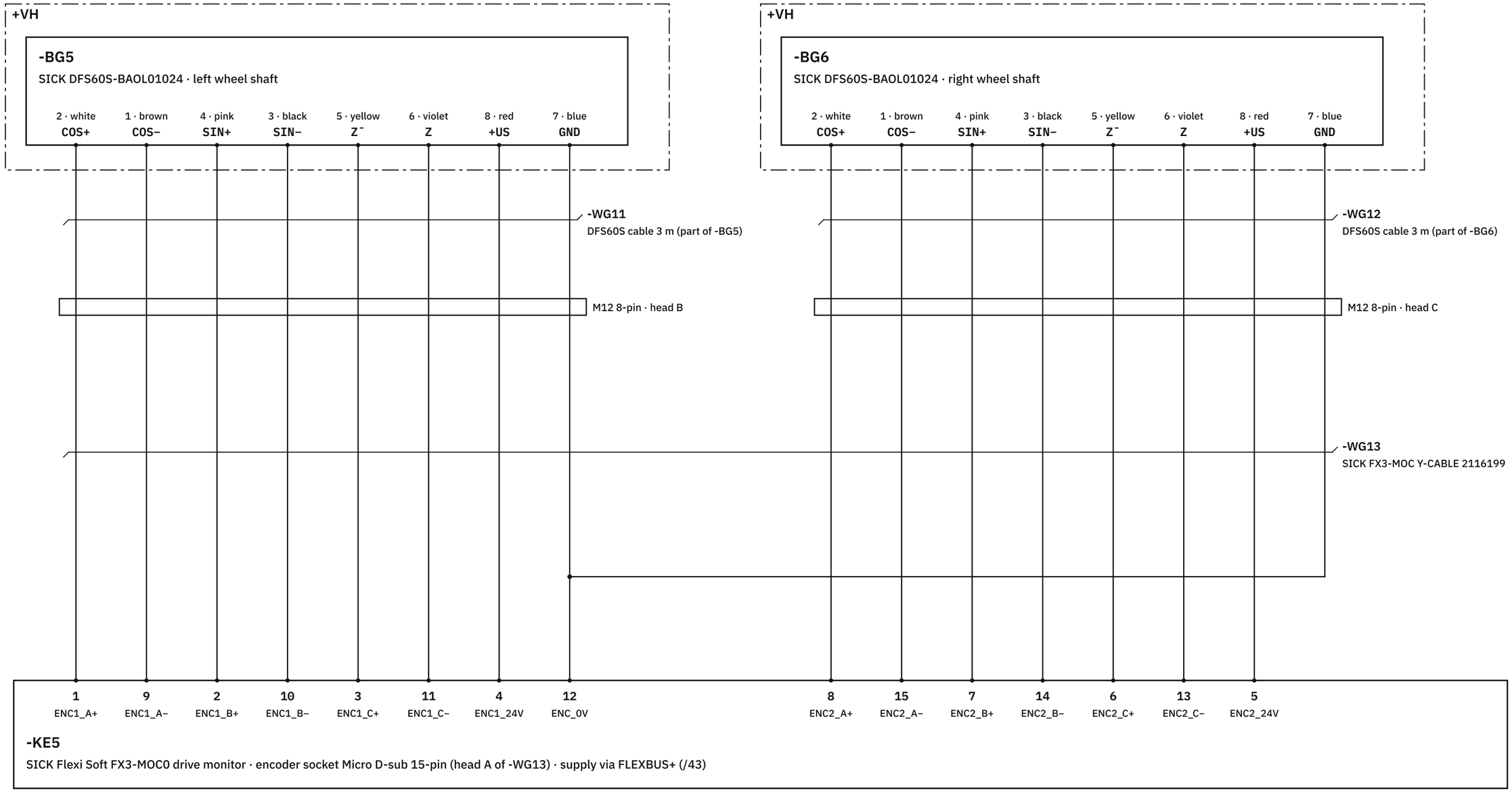
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Title: AGV I-PRIME driverless towing vehicle
Document type: Circuit diagram
Page description: Safety controller I/O table

Location: +EB
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 44
Number of pages: 24

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Notes

1. Speed limit in AUTO: above 1.1 m/s, -KE1 switches Q1, Q2 off (STO, /42).
2. Standstill monitoring: -KE5 checks that both wheels stand still while torque is off.
3. Speed-dependent scanner field switching: planned, not wired in this revision.
4. ENC1 = left wheel, ENC2 = right wheel. Encoder supply 24 V from -KE5.
5. Hollow shaft Ø6 mm, feather key, stator coupling: mount per the SICK instructions.
6. All 8 wires go through -WG13. Z¯, Z → ENC_C (pins 3, 11 / 6, 13): not evaluated in sin/cos mode.
7. Screens on the M12 and D-sub housings. Signals: sin/cos, 1 Vpp differential.
8. All devices in +EB unless boxed.

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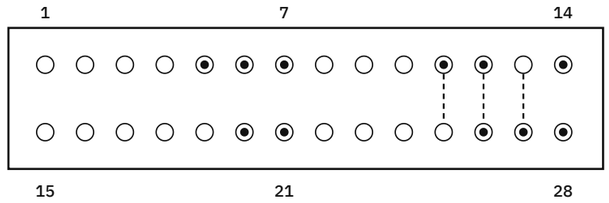
Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **Safety: wheel encoders and drive monitor**

Location: +EB +VH
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 45
Number of pages: 24

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BLVD-KRD control connector, 28 pins (same on -TA1 and -TA2), view on the driver



● = wired pin

--- = factory jumper, removed

Pin	Signal	Connected to	Wire marker	Page
1	OUT0+	—		
2	OUT0−	—		
3	OUT1+	—		
4	OUT1−	—		
5	CAN_L	CAN L	CAN L	51
6	CAN_H	CAN H	CAN H	51
7	CAN_GND	CAN GND	CAN GND	51
8	485GND	—		
9	TR+	—		
10	TR−	—		
11	HWTO1+	-KE2:Q1 (STO channel 1)	HWTO1	42
12	HWTO1−	1N24	1N24	42
13	0V	—		
14	EDM−	-KE2:I3 (left) / I4 (right)	EDM-L / EDM-R	42

Pin	Signal	Connected to	Wire marker	Page
15	IN-COM	—		
16	IN0	—		
17	IN1	—		
18	IN2	—		
19	IN3	—		
20	NET-VIN	1P24	1P24	31
21	NET-GND	1N24	1N24	31
22	485GND	—		
23	TR+	—		
24	TR−	—		
25	+V	—		
26	HWTO2+	-KE2:Q2 (STO channel 2)	HWTO2	42
27	HWTO2−	1N24	1N24	42
28	EDM+	1P24	1P24	42

Notes

1. Factory jumpers 11–25, 12–26 and 13–27 are removed on both drivers: the hardwired torque-off (HWTO) inputs are driven by the safety controller (page 42).
2. Grey pins are not connected. The drivers take no commands on their digital inputs; motion is commanded over CAN only (page 51).
3. Driver power: Oriental Motor cable LC03D06A (page 21). Motor and encoder: Oriental Motor cable CCM020B1AAF, 2 m (page 21).

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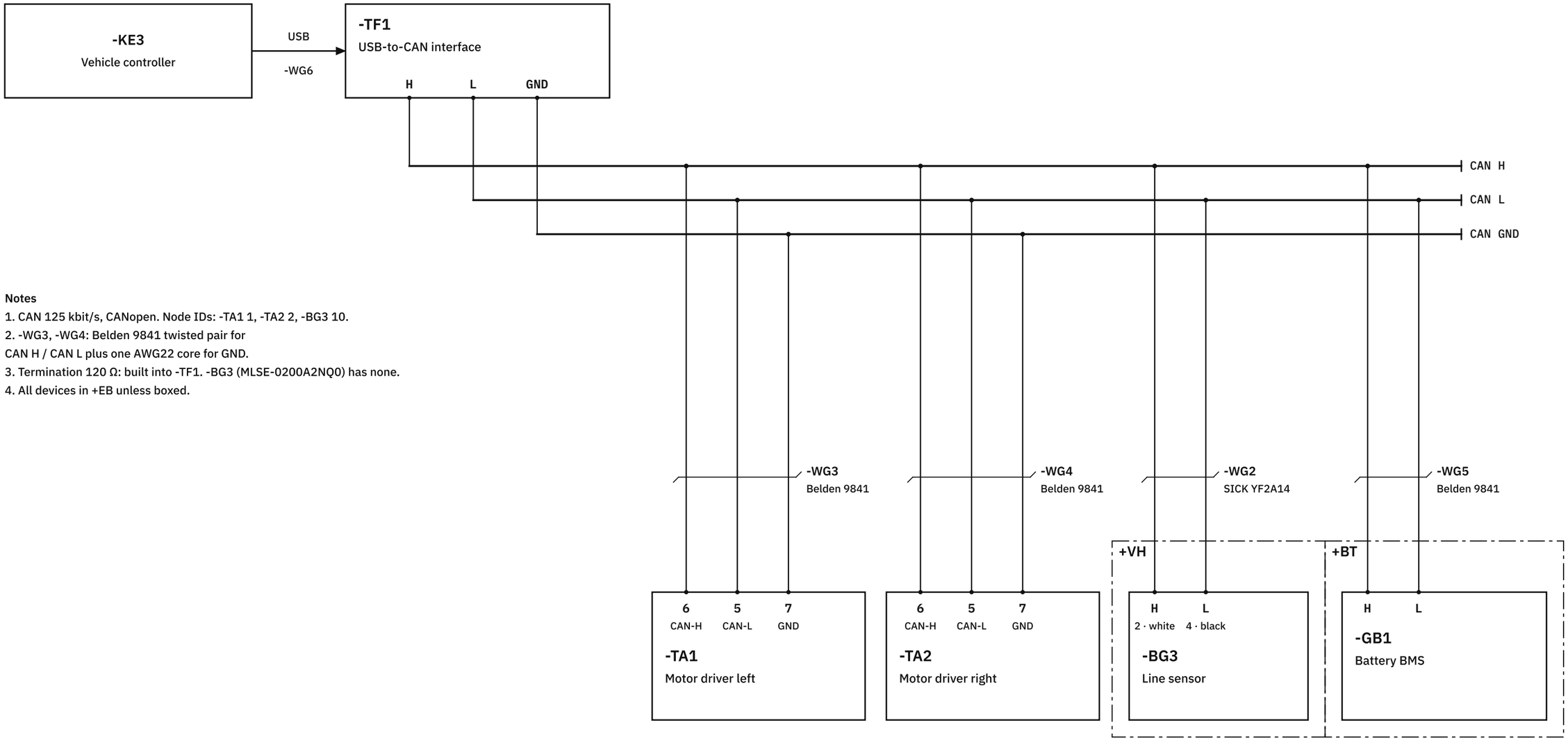
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Title: AGV I-PRIME driverless towing vehicle
Document type: Circuit diagram
Page description: Motor drivers: control connector

Location: +EB
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 50
Number of pages: 24



Notes

1. CAN 125 kbit/s, CANopen. Node IDs: -TA1 1, -TA2 2, -BG3 10.
2. -WG3, -WG4: Belden 9841 twisted pair for CAN H / CAN L plus one AWG22 core for GND.
3. Termination 120 Ω: built into -TF1. -BG3 (MLSE-0200A2NQ0) has none.
4. All devices in +EB unless boxed.

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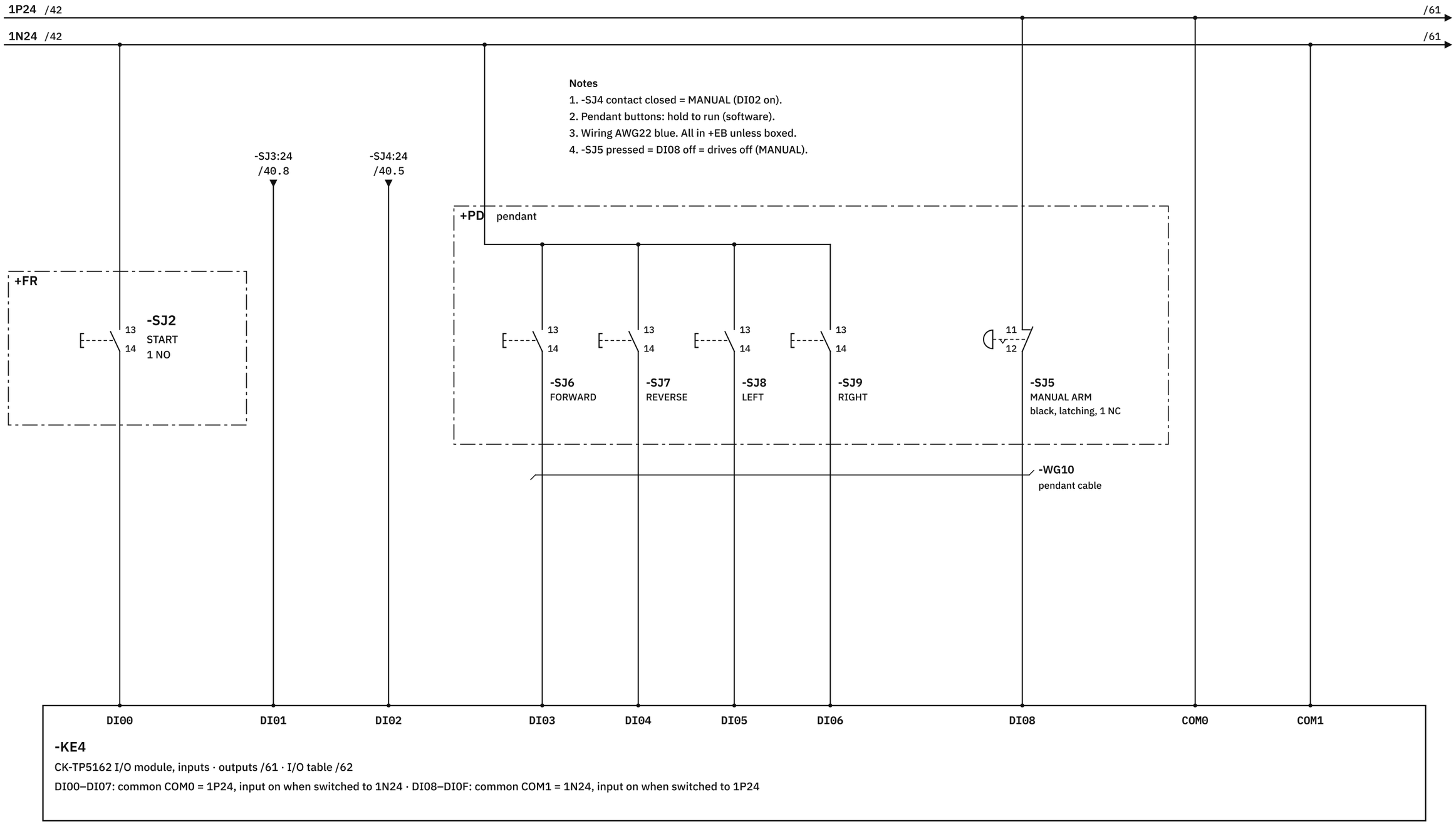
Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram

Page description: **CAN bus**

Location	+EB +VH
Language	en
Document number	IGP-IPR-CD-001
Revision	A
Page	51
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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **I/O module: inputs**

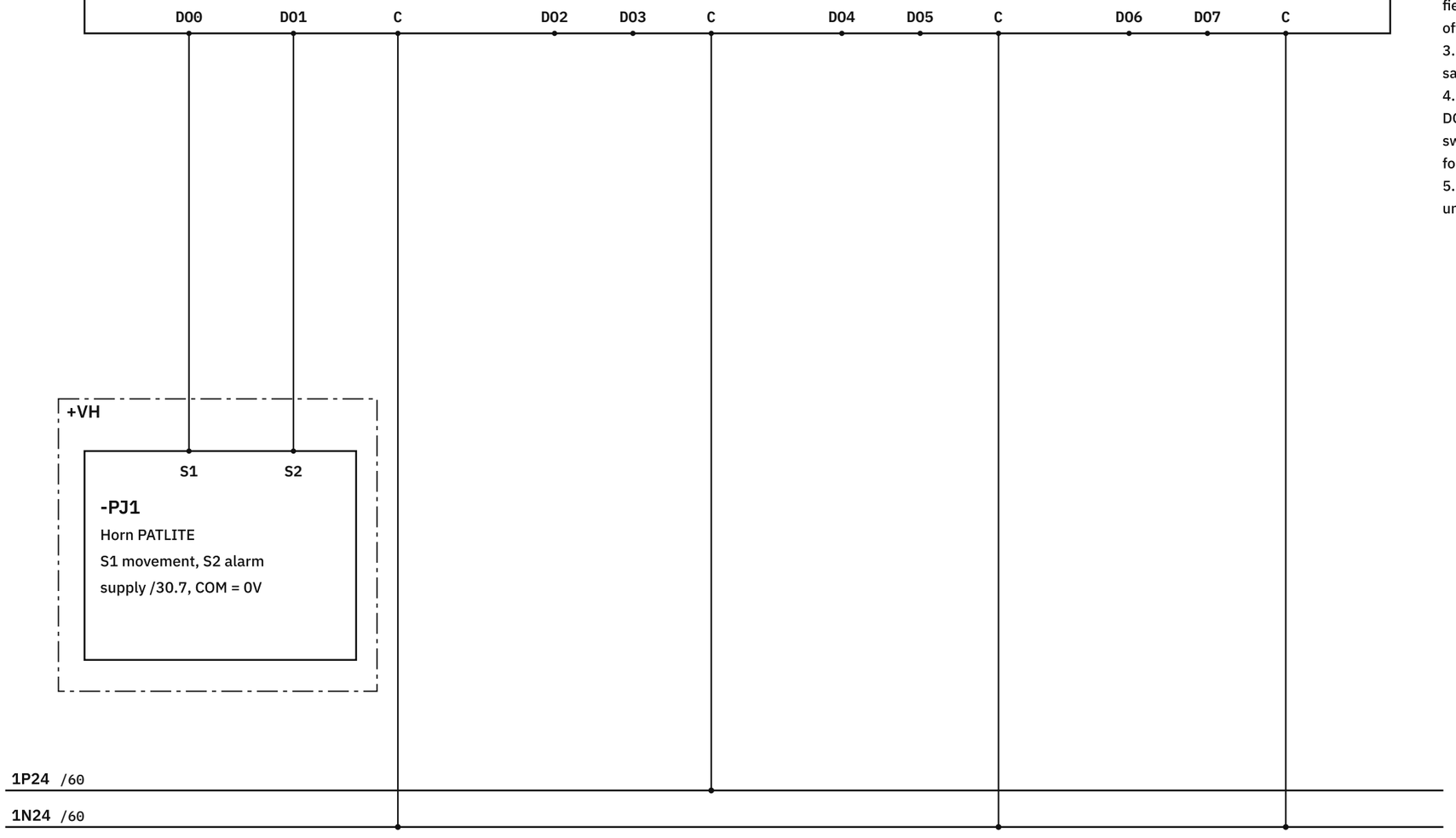
Location: +EB +FR +PD
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: 60
Number of pages: 24

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-KE4

CK-TP5162 I/O module, 16 outputs DO0…DOF: each output closes to the common C of its pair



Notes

1. DO0, sound 1 (movement): on while the wheels turn, in every mode.
2. DO1, sound 2 (alarm): 2 s after START before moving, and while a scanner field is occupied in AUTO. Sound 1 is off while sound 2 plays.
3. The horn is software-driven, not a safety function.
4. DO2 … DO7 spare; commons wired. DO8 … DOF not wired. The software also switches DO8 with motion: free output for a warning lamp.
5. Wiring AWG22 blue. All in +EB unless boxed.

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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram
Page description: **I/O module: outputs**

Location: +EB +VH
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: **61**
Number of pages: 24

-KE4 CK-TP5162 — I/O module, 16 DI / 16 DO, Ethernet (+EB)

Channel	Signal	Device	Function	Page
DI00	START	-SJ2	Start / continue automatic operation	60
DI01	RESET	-SJ3 (23-24)	Reset, second contact	60
DI02	MANUAL	-SJ4 (23-24)	ON = MANUAL, OFF = AUTO	60
DI03	P.FWD	-SJ6	Pendant forward (hold to run)	60
DI04	P.RVS	-SJ7	Pendant reverse (hold to run)	60
DI05	P.LFT	-SJ8	Pendant left (hold to run)	60
DI06	P.RGT	-SJ9	Pendant right (hold to run)	60
DI07			spare	
DI08	MANUAL ARM	-SJ5	On (button released) = drives on for manual driving	60
DI09–DI0F			spare	
DO0	HORN 1	-PJ1	Movement sound: on while the wheels turn	61
DO1	HORN 2	-PJ1	Alarm sound: 2 s after START, field occupied in AUTO	61
DO2–DO7			spare (commons wired)	61
DO8			switched with motion by the software; nothing connected	61
DO9–DOB			spare, not wired	

Input commons

DI00 … DI07: common COM0 = 1P24. An input is active when it is switched to 1N24.

DI08 … DI0F: common COM1 = 1N24. An input is active when it is switched to 1P24.

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Page description: I/O module I/O table

Location: +EB
Language: en

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Revision: A
Page: 62
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Cables: power (-WD) and control / data (-WG)

Tag	Type	Cores × size	From	To	Pages	Length
-WD1	NYAF single cores, black	2 × 4 mm²	-GB1	-QA1	20	
-WD2	NYAF single cores, black	2 × 2.5 mm²	-FC1, 1N48 bus	-WD6	21	
-WD3	NYAF single cores, black	2 × 2.5 mm²	-FC2, 1N48 bus	-WD7	21	
-WD4	NYAF single cores, black	2 × 1.5 mm²	1P48, 1N48	-TA3 in	20	
-WD5	NYAF single cores, black	2 × 1.5 mm²	1P48, 1N48	-TA4 in	20	
-WD6	Oriental Motor LC03D06A	3 cores (+, 0V, PE)	-WD2, PE	-TA1 power	21	
-WD7	Oriental Motor LC03D06A	3 cores (+, 0V, PE)	-WD3, PE	-TA2 power	21	
-WD8	Oriental Motor CCM020B1AAF	motor + encoder	-TA1	-MA1	21	2 m
-WD9	Oriental Motor CCM020B1AAF	motor + encoder	-TA2	-MA2	21	2 m
-WD10	Display power cable	2 cores	1P24, 1N24	-PH1	31	
-WD11	IPC power cable	2 cores	1P12, 1N12	-KE3	32	
-WD12	RFID power cable	2 cores	1P12, 1N12	-BG4	32	
-WG1	SICK YF2A18-050UA5XLEAX, M12 8-pin female → open end, PUR	8 × 0.25 mm²	1P24, 1N24; -KE2:I5, I6	-BG1	30, 41	5 m
-WG2	SICK YF2A14-020UB3XLEAX, M12 4-pin female → open end, PUR	4 cores	1P24, 1N24; CAN bus	-BG3	31, 51	2 m
-WG3	Belden 9841 (CAN H, L) + AWG22 (GND)	1 × 2 × 0.32 mm² + 1	CAN bus	-TA1:5, 6, 7	51	
-WG4	Belden 9841 (CAN H, L) + AWG22 (GND)	1 × 2 × 0.32 mm² + 1	CAN bus	-TA2:5, 6, 7	51	
-WG5	Belden 9841 + battery comms cable	1 × 2 × 0.32 mm² + 1	CAN bus	-GB1 BMS	51	
-WG6	USB cable	USB	-KE3	-TF1	11, 51	
-WG7	Ethernet patch cable	RJ45	-KE3 enp2s0	-KE4 IN	11	
-WG8	Ethernet patch cable	RJ45	-KE4 OUT	-BG4	11	
-WG9	SICK YM2D24-020PN1MRJA4, Ethernet	M12 4-pin D-coded → RJ45	-KE3 enp1s0	-BG1	11, 41	2 m
-WG10	Pendant cable (Delta)	8 × 0.5 mm²	-KE4:DI03–DI06, DI08; 1N24, 1P24	+PD	60	
-WG11	SICK DFS60S cable (part of -BG5), M12 plug fitted, all 8 wires used	8 × 0.15 mm², screened	-BG5	-WG13 head B	45	3 m
-WG12	SICK DFS60S cable (part of -BG6), M12 plug fitted, all 8 wires used	8 × 0.15 mm², screened	-BG6	-WG13 head C	45	3 m
-WG13	SICK FX3-MOC Y-CABLE 2116199	AWG28 / AWG26, screened	-KE5 encoder socket	-WG11, -WG12	45	0.3 m

Notes

1. Single-core panel wiring is not listed: AWG22 blue (control, safety) and NYAF black (48 V) inside +EB and to the front panel +FR. It is identified by the potential or signal name on each wire marker.
2. Blank length: to be measured on the built vehicle.

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Page description: **Cable overview**

Location:
Language: en

Document number: IGP-IPR-CD-001
Revision: A
Page: **75**
Number of pages: 24

Parts list: all devices shown in this circuit diagram

Tag	Description	Maker	Part number	Rating	Location	Pages	Safety
-GB1	Battery with battery management system (BMS on CAN)	Incoe	48 V 50 Ah LFP	48 V, 50 Ah, $LiFePO_4$	+BT	20, 51	
-QA1	Circuit breaker, supply disconnecting device (main switch)	ABB	S202-C32	2P 32 A	+EB	20	
-FC1	Circuit breaker, feeder left motor driver	ABB	S201-C13	1P 13 A	+EB	20	
-FC2	Circuit breaker, feeder right motor driver	ABB	S201-C13	1P 13 A	+EB	20	
-FC3	Circuit breaker, 24 V bus	ABB	S201-C6	1P 6 A	+EB	30	
-FC4	Circuit breaker, 12 V bus	ABB	S201-C10	1P 10 A	+EB	32	
-TA1	Motor driver, left (CANopen, HWTO)	Oriental Motor	BLVD-KRD	48 V DC, 400 W	+EB	21, 42, 50, 51	Y
-TA2	Motor driver, right (CANopen, HWTO)	Oriental Motor	BLVD-KRD	48 V DC, 400 W	+EB	21, 42, 50, 51	Y
-TA3	DC/DC converter 48 → 24 V, isolated	Mean Well	SD-150C-24	24 V 6.25 A (150 W)	+EB	20, 30	
-TA4	DC/DC converter 48 → 12 V, isolated	Mean Well	SD-100C-12	12 V 8.3 A (100 W)	+EB	20, 32	
-MA1	Motor with holding brake, left; gearhead GFS6G30FR 1:30	Oriental Motor	BLMR6400SKM-GFV-B	400 W	+VH	21	
-MA2	Motor with holding brake, right; gearhead GFS6G30FR 1:30	Oriental Motor	BLMR6400SKM-GFV-B	400 W	+VH	21	
-KE1	Safety controller main module, with system plug FX3-MPL000001	SICK	FX3-CPU000000	24 V DC	+EB	43	Y
-KE2	Safety controller I/O module, 8 in / 4 out	SICK	FX3-XTIO84002	24 V DC	+EB	40–44	Y
-KE5	Safety controller drive monitor, 2 encoder inputs	SICK	FX3-MOC000000	via FLEXBUS+	+EB	12, 43, 45	Y
-KE3	Vehicle controller, industrial PC (Intel N97)	NEXCOM	NEU-X102-N97	12 V DC	+EB	11, 32, 51	
-KE4	I/O module, 16 DI / 16 DO, Ethernet	CKDZ-IoT	CK-TP5162	24 V DC	+EB	30, 60–62	
-TF1	USB-to-CAN interface	CANable	CANable V2.0S	USB	+EB	11, 51	
-SJ1	Emergency stop push-button, 2 NC, turn to release	Hanyoung Nux	ARE-3-R-22-R		+FR	40	Y
-SJ2	Push-button START, green, 1 NO	Hanyoung Nux	ARF-F-1-10-G		+FR	60	
-SJ3	Push-button RESET, yellow, 2 NO	Hanyoung Nux	ARF-F-1-20-Y		+FR	40, 60	Y
-SJ4	Selector switch MANUAL / AUTO, 2 positions, 2 NO	Hanyoung Nux	ARS-S-2-12		+FR	40, 60	Y
-SJ5	Push-button MANUAL ARM, mushroom, black, latching, turn to release, 1 NC (not an emergency stop)	Hanyoung Nux	HY-PA04		+PD	60	
-SJ6	Pendant button FORWARD, 1 NO	Hanyoung Nux	HY-PA04		+PD	60	
-SJ7	Pendant button REVERSE, 1 NO	Hanyoung Nux	HY-PA04		+PD	60	
-SJ8	Pendant button LEFT, 1 NO	Hanyoung Nux	HY-PA04		+PD	60	
-SJ9	Pendant button RIGHT, 1 NO	Hanyoung Nux	HY-PA04		+PD	60	
-BG1	Safety laser scanner, front; system plug NANSX-AAABAEZZ1	SICK	NANS3-AAAZ30AN1	24 V DC	+VH	30, 41	Y
-BG3	Magnetic line sensor with IMU (CANopen), M12 4-pin lead 0.3 m	SICK	MLSE-0200A2NQ0	24 V DC	+VH	31, 51	
-BG4	UHF RFID reader (Ethernet)	Chafon	CF821	12 V DC	+VH	11, 32	
-BG5	Safety encoder, left wheel shaft (sin/cos, 1 024 periods)	SICK	DFS60S-BAOL01024	24 V DC from -KE5	+VH	12, 45	Y
-BG6	Safety encoder, right wheel shaft (sin/cos, 1 024 periods)	SICK	DFS60S-BAOL01024	24 V DC from -KE5	+VH	12, 45	Y
-PJ1	Melody horn, 2 sounds	PATLITE	BK-24AE-K	24 V DC	+VH	30, 61	
-PH1	Display	Jawest	JWS080-M-QD	24 V DC	+FR	31	
-GQ1	Fan	NMB-MAT	3110KL-05W-B60	24 V DC	+EB	31	
-GQ2	Fan	NMB-MAT	3110KL-05W-B60	24 V DC	+EB	31	
-GQ3	Fan	NMB-MAT	3110KL-05W-B60	24 V DC	+EB	31	
-XE1	Chassis earth point (PE)				+EB	21	

Notes

Safety = Y: the part is in the safety chain (page 12).

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

Rev	Change	Date	Name
A	First issue, draft	2026-11-01	JOY M.

Legal owner

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DRAFT — for preliminary review — not for production